



| Accession           | Protein                | Sequence         | Position |
|---------------------|------------------------|------------------|----------|
| Varen 6869          | AgSp1.2 glycine rm1 1  | PGTTPGTTTRHDKGPK | 1        |
| Varen 6869          | AgSp1.1 glycine rm1 1  | PGTTPGTTTRHDKGPK | 1        |
| Varen 6869          | AgSp1.2 glycine rm1 2  | PGTTPGTTTRHDKGPK | 2        |
| Varen 6869          | AgSp1.1 glycine rm1 2  | PGTTPGTTTRHDKGPK | 2        |
| Varen 6869          | AgSp1.2 glycine rm1 3  | PGTTPGTTTRHDKGPK | 3        |
| Varen 6869          | AgSp1.2 glycine rm1 10 | PGTTPGTTTRHDKGPK | 10       |
| Varen 6869          | AgSp1.1 glycine rm1 11 | PGTTPGTTTRHDKGPK | 11       |
| Varen 6869          | AgSp1.2 glycine rm1 11 | PGTTPGTTTRHDKGPK | 11       |
| Varen 6869          | AgSp1.2 glycine rm1 12 | PGTTPGTTTRHDKGPK | 12       |
| Varen 6869          | AgSp1.2 glycine rm1 13 | PGTTPGTTTRHDKGPK | 13       |
| Varen 6869          | AgSp1.2 glycine rm1 14 | PGTTPGTTTRHDKGPK | 14       |
| MLaby 7048          | AgSp1 glycine rm1b 2   | PGTTPGTTTRHDKGPK | 2        |
| MLaby 7048          | AgSp1 glycine rm1a 2   | PGTTPGTTTRHDKGPK | 2        |
| MLaby 7048          | AgSp1 glycine rm1a 4   | PGTTPGTTTRHDKGPK | 4        |
| MLaby 7048          | AgSp1 glycine rm1a 13  | PGTTPGTTTRHDKGPK | 13       |
| MLaby 7048          | AgSp1 glycine rm2a 21  | PGTTPGTTTRHDKGPK | 21       |
| MLaby 7048          | AgSp1 glycine rm2a 22  | PGTTPGTTTRHDKGPK | 22       |
| MLaby 7048          | AgSp1 glycine rm2b 29  | PGTTPGTTTRHDKGPK | 29       |
| Msag 18528          | AgSp1.2 glycine rm 1   | PGTTPGTTTRHDKGPK | 1        |
| Msag 18528          | AgSp1.2 glycine rm 16  | PGTTPGTTTRHDKGPK | 16       |
| Msag 18528          | AgSp1.2 glycine rm 112 | PGTTPGTTTRHDKGPK | 112      |
| Msag 18528          | AgSp1.2 glycine rm 113 | PGTTPGTTTRHDKGPK | 113      |
| Msag 18528          | AgSp1.2 glycine rm 114 | PGTTPGTTTRHDKGPK | 114      |
| Msag 18528          | AgSp1.2 glycine rm 115 | PGTTPGTTTRHDKGPK | 115      |
| Ncruc 58532and13532 | AgSp1 glycine rm 1     | PGTTPGTTTRHDKGPK | 1        |
| Ncruc 58532and13532 | AgSp1 glycine rm 2     | PGTTPGTTTRHDKGPK | 2        |
| Ncruc 58532and13532 | AgSp1 glycine rm 3     | PGTTPGTTTRHDKGPK | 3        |
| Ncruc 58532and13532 | AgSp1 glycine rm 4     | PGTTPGTTTRHDKGPK | 4        |
| Ncruc 58532and13532 | AgSp1 glycine rm 5     | PGTTPGTTTRHDKGPK | 5        |
| Ncruc 58532and13532 | AgSp1 glycine rm 6     | PGTTPGTTTRHDKGPK | 6        |
| Lcorn 18104         | AgSp1.1 glycine rm1 1  | PGTTPGTTTRHDKGPK | 1        |
| Lcorn 18104         | AgSp1.1 glycine rm1 37 | PGTTPGTTTRHDKGPK | 37       |
| Lcorn 18104         | AgSp1.1 glycine rm1 38 | PGTTPGTTTRHDKGPK | 38       |
| Lcorn 18104         | AgSp1.1 glycine rm1 39 | PGTTPGTTTRHDKGPK | 39       |
| Atri AgSp1 genomic  | glycine rm1 1          | PGTTPGTTTRHDKGPK | 1        |
| Atri AgSp1 genomic  | glycine rm1 2          | PGTTPGTTTRHDKGPK | 2        |
| Atri AgSp1 genomic  | glycine rm1 47         | PGTTPGTTTRHDKGPK | 47       |
| Atri AgSp1 genomic  | glycine rm2 40         | PGTTPGTTTRHDKGPK | 40       |
| Amarm 50219         | AgSp1.1 glycine rm1 1  | PGTTPGTTTRHDKGPK | 1        |
| Amarm 50219         | AgSp1.1 glycine rm1 2  | PGTTPGTTTRHDKGPK | 2        |
| Amarm 50219         | AgSp1.1 glycine rm1 3  | PGTTPGTTTRHDKGPK | 3        |
| Amarm 50219         | AgSp1.1 glycine rm1 8  | PGTTPGTTTRHDKGPK | 8        |
| Amarm 50219         | AgSp1.1 glycine rm2 1  | PGTTPGTTTRHDKGPK | 1        |
| Amarm 50219         | AgSp1.1 glycine rm2 78 | PGTTPGTTTRHDKGPK | 78       |
| Amarm 50219         | AgSp1.1 glycine rm2 89 | PGTTPGTTTRHDKGPK | 89       |
| Aarg AgSp1          | FC511 glycine rm1 1    | PGTTPGTTTRHDKGPK | 1        |
| Aarg AgSp1          | FC511 glycine rm1 2    | PGTTPGTTTRHDKGPK | 2        |
| Aarg AgSp1          | FC511 glycine rm1 3    | PGTTPGTTTRHDKGPK | 3        |
| Aarg AgSp1          | FC511 glycine rm1 13   | PGTTPGTTTRHDKGPK | 13       |
| Aarg AgSp1          | FC511 glycine rm1 15   | PGTTPGTTTRHDKGPK | 15       |
| Aarg AgSp1          | FC511 glycine rm2 6    | PGTTPGTTTRHDKGPK | 6        |
| Aarg AgSp1          | FC511 glycine rm2 7    | PGTTPGTTTRHDKGPK | 7        |
| Mhut 1271           | AgSp1 glycine rm1 1    | PGTTPGTTTRHDKGPK | 1        |
| Mhut 1271           | AgSp1 glycine rm1 2    | PGTTPGTTTRHDKGPK | 2        |
| Mhut 1271           | AgSp1 glycine rm1 3    | PGTTPGTTTRHDKGPK | 3        |
| Mhut 1271           | AgSp1 glycine rm1 4    | PGTTPGTTTRHDKGPK | 4        |
| Mhut 1271           | AgSp1 glycine rm1 5    | PGTTPGTTTRHDKGPK | 5        |
| Mhut 1271           | AgSp1 glycine rm1 6    | PGTTPGTTTRHDKGPK | 6        |
| Mhut 1271           | AgSp1 glycine rm1 7    | PGTTPGTTTRHDKGPK | 7        |
| Mhut 1271           | AgSp1 glycine rm1 8    | PGTTPGTTTRHDKGPK | 8        |
| Mhut 1271           | AgSp1 glycine rm3 17   | PGTTPGTTTRHDKGPK | 17       |

|                                        |                                                                                  |     |
|----------------------------------------|----------------------------------------------------------------------------------|-----|
| Varen 6869 AgSp1.2 glycine rm1 1       | .....PETHPA <b>GRPIQIV</b> PAG                                                   | 86  |
| Varen 6869 AgSp1.1 glycine rm1 1       | .....GPGKMFGPMRPRRQRTTPPS <b>VEGPNCKPIQIE</b> PAG                                | 92  |
| Varen 6869 AgSp1.2 glycine rm1 2       | .....QGM...NFATTT <b>PIPGGGRPIQIE</b> PAG                                        | 99  |
| Varen 6869 AgSp1.1 glycine rm1 2       | GGY..PMRPR.....                                                                  | 83  |
| Varen 6869 AgSp1.2 glycine rm1 3       | .....                                                                            | 11  |
| Varen 6869 AgSp1.2 glycine rm1 10      | .....HMGSDEQGMNFVTT <b>TPIPGGGRPIQIE</b> PAG                                     | 37  |
| Varen 6869 AgSp1.1 glycine rm1 11      | .....GPEG.....PGGYPMGPGNPFPRPMGPGRQRTTPSS <b>VRGPKCKPIQIE</b> PAG                | 45  |
| Varen 6869 AgSp1.2 glycine rm1 11      | .....QGM...NFATTT <b>PIPGGGRPIQIE</b> PAG                                        | 99  |
| Varen 6869 AgSp1.2 glycine rm1 12      | .....QGM...NFATTT <b>PIPGGGRPIQIE</b> PAG                                        | 99  |
| Varen 6869 AgSp1.2 glycine rm1 13      | .....QGI...NFATTT <b>PIPGGGRPIQIE</b> PAG                                        | 99  |
| Varen 6869 AgSp1.2 glycine rm1 14      | .....                                                                            | 45  |
| Mlaby 7048 AgSp1 glycine rm1b 2        | .....TGPGGQPLNPQ..GPISPFPGGGQSM...NPLGPGGS..GG...QTT <b>TPEPIPGPNCKPLQIV</b> PAG | 128 |
| Mlaby 7048 AgSp1 glycine rm1a 2        | .F.....Y.....FP.....VT...PK <b>PTPTPVTPGGCKPIQII</b> PAG                         | 101 |
| Mlaby 7048 AgSp1 glycine rm1a 4        | .....                                                                            | 53  |
| Mlaby 7048 AgSp1 glycine rm1a 13       | .....QFPTGPGGQPLNPT..G.....LGPRGQPL.....NPLGPGSPGGQTT <b>TAPVPGPDCKPLQIV</b> PAG | 91  |
| Mlaby 7048 AgSp1 glycine rm2a 21       | .....PTGPGGQPLNP.....LGPRGQPL.....NPLGPGSPGGQTT <b>TAPVPGPDCKPLQIV</b> PAG       | 60  |
| Mlaby 7048 AgSp1 glycine rm2a 22       | .....                                                                            | 16  |
| Mlaby 7048 AgSp1 glycine rm2b 29       | .....                                                                            | 23  |
| Msag 18528 AgSp1.2 glycine rm 1        | .....                                                                            | 16  |
| Msag 18528 AgSp1.2 glycine rm 16       | .....GPQGGPGGYPLGPGGQ.....T.....                                                 | 16  |
| Msag 18528 AgSp1.2 glycine rm 112      | .....RGPQGPGGYPLGPGSPFPGPKGPGGQTT <b>TASVQGPDCGEPIQVE</b> PAG                    | 46  |
| Msag 18528 AgSp1.2 glycine rm 113      | GG.....YPLGPGSPFPGPKGPGGQTT <b>TASVQGPDCGEPIQVE</b> PAG                          | 115 |
| Msag 18528 AgSp1.2 glycine rm 114      | .....                                                                            | 70  |
| Msag 18528 AgSp1.2 glycine rm 115      | .....                                                                            | 35  |
| Ncruc 58532andi3532 AgSp1 glycine rm 1 | .....PGFNF....GTPAT <b>TPTPIQPGCKPIQVV</b> PAG                                   | 103 |
| Ncruc 58532andi3532 AgSp1 glycine rm 2 | .....PGFNF....GTPAT <b>TPTPIQPGCKPIQVV</b> PAG                                   | 103 |
| Ncruc 58532andi3532 AgSp1 glycine rm 3 | .....PGFNF....GTPAT <b>TPTPIQPGCKPIQVV</b> PAG                                   | 103 |
| Ncruc 58532andi3532 AgSp1 glycine rm 4 | .....PGFNF....GTPAT <b>TPTPIQPGCKPIQVV</b> PAG                                   | 103 |
| Ncruc 58532andi3532 AgSp1 glycine rm 5 | .....PGFNF....GTPAT <b>TPTPIQPGCKPIQVV</b> PAG                                   | 103 |
| Ncruc 58532andi3532 AgSp1 glycine rm 6 | .....                                                                            | 59  |
| Lcorn 18104 AgSp1.1 glycine rm1 1      | GI.....NPYYPMRPGR.....                                                           | 87  |
| Lcorn 18104 AgSp1.1 glycine rm1 37     | GL.....NPYYPMGPEGKPM...YPYGPSPFGPQGPGGQT <b>TKTPVPGPDCKPIQIV</b> PAG             | 126 |
| Lcorn 18104 AgSp1.1 glycine rm1 38     | GQ.....PL.....FPFGPQ...GPV...GQT <b>TTTPIPGPDCKPIQIV</b> PAG                     | 110 |
| Lcorn 18104 AgSp1.1 glycine rm1 39     | .....                                                                            | 45  |
| Atri AgSp1 genomic glycine rm1 1       | .....PKTPQGSQGQPM...YPFGPGSPYGP...EQT <b>TTTPIPGPDCKPIQIE</b> PAG                | 114 |
| Atri AgSp1 genomic glycine rm1 2       | GG.....MPL..SPYSPQGPGGQPM...YPFGPGSPYGP...EQT <b>TTTPIPGPDCKPIQIE</b> PAG        | 116 |
| Atri AgSp1 genomic glycine rm1 47      | .....                                                                            | 13  |
| Atri AgSp1 genomic glycine rm2 40      | .....KGPLGPGGQPM...YPSGPQGTG...GQT <b>TTTPIPGPDCKPIQIE</b> PAG                   | 97  |
| Amarm 50219 AgSp1.1 glycine rm1 1      | GG.....                                                                          | 98  |
| Amarm 50219 AgSp1.1 glycine rm1 2      | GG.....QTTPT <b>TPVPGPRCKPIQIL</b> PAG                                           | 88  |
| Amarm 50219 AgSp1.1 glycine rm1 3      | .....                                                                            | 10  |
| Amarm 50219 AgSp1.1 glycine rm1 8      | GF.....NFA..TP.....ATT <b>PTPIPGPGCKPIQIV</b> PAG                                | 103 |
| Amarm 50219 AgSp1.1 glycine rm2 1      | GG.....FPLGP...AGFPLGPGGQPI...YPYGPSPYGPQGPGGKT <b>TTTPIPGPDCKPIQIV</b> PAG      | 131 |
| Amarm 50219 AgSp1.1 glycine rm2 78     | .....APTGGGGFPLRT.....TGPGGQPI...YPYGPSPYGPQGPGGK.....                           | 37  |
| Amarm 50219 AgSp1.1 glycine rm2 89     | .....APTGGGGFPL.....PGPGGQPI...YPYGPSPYGPQGPGGK.....                             | 35  |
| Aarg AgSp1 FC511 glycine rm1 1         | .....PEEPSGRPM...YPFGPGSPYGP...EQT <b>TTTPIPGPDCKPIQIE</b> PAG                   | 114 |
| Aarg AgSp1 FC511 glycine rm1 2         | GG.....MPL..SPYYPPQGPGGQPM...NPFAPGSPYGP...EQT <b>TTTPIPGPDCKPIQIE</b> PAG       | 129 |
| Aarg AgSp1 FC511 glycine rm1 3         | .....                                                                            | 22  |
| Aarg AgSp1 FC511 glycine rm1 13        | DG.....QPMYPS..GP.....Q...GPG.GQT <b>TTTPIPGPDCKPIQIE</b> PAG                    | 91  |
| Aarg AgSp1 FC511 glycine rm1 15        | .....                                                                            | 62  |
| Aarg AgSp1 FC511 glycine rm2 6         | .....IPT <b>TTTPIPGPGGQPIQII</b> PAG                                             | 96  |
| Aarg AgSp1 FC511 glycine rm2 7         | .....                                                                            | 45  |
| Mhut 1271 AgSp1 glycine rm1 1          | K.....RPSY..THVKD <b>KPIQIV</b> PAG                                              | 96  |
| Mhut 1271 AgSp1 glycine rm1 2          | .....LGK <b>ASHCKPIQII</b> PAG                                                   | 92  |
| Mhut 1271 AgSp1 glycine rm1 3          | GFYFG.....MPQY <b>IPGGGRPIQIV</b> PAG                                            | 99  |
| Mhut 1271 AgSp1 glycine rm1 4          | GFYFG.....MPQY <b>IPGGGRPIQIV</b> PAG                                            | 99  |
| Mhut 1271 AgSp1 glycine rm1 5          | GFYFG.....MPQY <b>IPGGCKPIQIE</b> PAG                                            | 96  |
| Mhut 1271 AgSp1 glycine rm1 6          | GFYFG.....MPQY <b>IPGGCKPIQIE</b> PAG                                            | 99  |
| Mhut 1271 AgSp1 glycine rm1 7          | GFYFG.....MPQY <b>IPGGCKPIQIE</b> PAG                                            | 99  |
| Mhut 1271 AgSp1 glycine rm1 8          | GFYFG.....MPQY <b>IPGGCKPIQIV</b> PAG                                            | 99  |
| Mhut 1271 AgSp1 glycine rm3 17         | .....                                                                            | 45  |
| Mhut 1271 AgSp1 glycine rm3 17         | .....                                                                            | 45  |

 non-conserved  
 similar  
  $\geq 50\%$  conserved